Closing the central-window memory Gram
A polarization proof of the output–force cross

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Abstract

The remaining central-window cross can be obtained from adjacent-charge polarization and a neutral moment, without assuming the candidate charged Casimir formula. A normalization cancellation improves the neutral moment's one-step drift to $O(1/(Np))$. The resulting cross equals the output-deficit self norm in the double-scaled limit. This closes the four-field equal-time Gram and extends the positive projected quadratic memory-core variance to compact central charge windows. Energy-resolved insertions are the next step.

1 Operators and the result

Use the full overlapping complex Gaussian p-body ensemble, common couplings across charges, normalized maximally mixed sector trace, and $\hbar = 1$. Let N be even, $p^2/N = \lambda_N \to \lambda > 0$, and $m = N/2 + xp$, $|x| \le L$. All bounds are uniform on this window and positive compact λ_N intervals. Set

$$C_m = \mathbb{E}H_m^2 = w_N \binom{m}{p}\binom{N-m+p}{p}, \qquad q = e^{-4\lambda}, \tag{1}$$

$$B = \mathsf{W}_2 H_m^2, \qquad O = \mathsf{W}_2[\mathrm{Rem}(H_{m-1}^2) + (2p-m)H_m^2], \tag{2}$$

$$L_{\mathrm{in}} = \mathsf{W}_2\{h, D - \bar{u}h\}, \qquad D = \sum_a (p-a)h_a, \tag{3}$$

$$\mathcal{S} = \sum_{a \le b, r} (a+b-r)\mathsf{W}_2 Z_{ab;r}^m, \qquad \bar{u} = \sum_a (p-a)C_a/C_m. \tag{4}$$

Here W_2 is ordinary second Gaussian chaos, $Z_{ab;r}^m = P_r(h_a \odot h_b)$, and $\odot$ means square for $a = b$ and anticommutator otherwise. Put

$$z = 1+q, \quad b = 2\lambda(1-q), \quad l = 2\lambda(1+q), \quad s = 4\lambda^2(1+q)+2\lambda(1-q), \quad a_* = 4\lambda+4\lambda^2(1+q). \tag{5}$$

The new entry and the resulting complete Gram are

$$\langle \mathcal{S}, O\rangle / C_m^2 \to s, \qquad \frac{G(B, L_{\mathrm{in}}, \mathcal{S}, O)}{C_m^2} \to \begin{pmatrix} z & 0 & b & b \\ 0 & l & 0 & 0 \\ b & 0 & s & s \\ b & 0 & s & a_* \end{pmatrix}. \tag{6}$$

All other entries were established in the two preceding central notes. Inner products are $\mathbb{E}\,\mathrm{tr}_m(U^\dagger V)$.

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2 An exact polarization of the actual force

Use one common Gaussian coefficient Hilbert space, split into orthogonal unordered input pairs and output ranks. For each block, write $u = p-a$, $v = p-b$, $S = a+b-r$, $k = 2p-r = u+v+S$, and

$$U = \mathsf{W}_2 Z^m_{ab;r}/C_m, \qquad V = \mathcal{I}_m(\mathsf{W}_2 Z^{m-1}_{ab;r}/C_{m-1}), \qquad E = V - U, \tag{7}$$

$$t_r = R_m \ell_r, \qquad R_m = \frac{(m-p)(n+p)}{mn}, \qquad n = N - m + 1, \tag{8}$$

$$\ell_r^2 = \frac{m}{n}[mn - r(N+1-r)], \qquad h_0 = m - 2p. \tag{9}$$

The preceding removal isometry $\mathcal{I}_m = \sum \ell_r^{-1} \operatorname{Rem} P_r$ acts on the common ranks $r \le 2p$. For sufficiently large p these ranks exist in both sectors. The exact force identity is

$$O_{ab;r}/C_m = t_r V - h_0 U. \tag{10}$$

For any real block weight W, polarization therefore gives

$$\begin{aligned}\sum W\langle U, O_{ab;r}/C_m\rangle = \frac{1}{2}\sum W t_r(\|V\|^2 - \|U\|^2) \\ + \sum W(t_r - h_0)\|U\|^2 - \frac{1}{2}\sum W t_r \|E\|^2.\end{aligned} \tag{11}$$

The inner products are real: each Gaussian coefficient of the symmetrized product is Hermitian, and removal preserves this property. Thus there is no phase approximation in (11).

Choose the charge-independent label weight

$$W = X/N, \quad X = x_{a+b} - x_r = S[N+1-2(a+b)+S], \quad x_j = j(N+1-j). \tag{12}$$

The earlier positive sixth output moment and factorial input tails, together with the coherent-force bound, give

$$\sum \|E\|^2 \le K/N^2, \quad \sum S^6\|U\|^2 + \sum S^6\|V\|^2 \le K, \quad 0 \le W \le Kp. \tag{13}$$

Fixed polynomial input weights can be inserted in the product bound. Uniform scalar expansion gives, for $0 \le r \le 2p$,

$$t_r \le KN, \quad |t_r - h_0| \le K(1+k), \quad |t_r - h_0 - (k - 2\lambda_N)| \le K(1+k)/p. \tag{14}$$

The last estimate follows from the uniform expansion at $r = \rho p$: $t_r - (m-r) = 2\lambda_N[x(2-\rho) - 1] + O(p^{-1})$ and $2 - \rho = k/p$. No restriction to bounded k is made here.

3 A small neutral drift after normalization

Define the known neutral first moment

$$F_m = \sum W \|\mathsf{W}_2 Z^m_{ab;r}/C_m\|^2 = \sum_{a,b} \omega_a^{(m)} \omega_b^{(m)} f_{ab}(m), \tag{15}$$

$$f_{ab}(m) = \frac{1}{N}\left[-2ab(1+\chi_{ab}^{(m)}) + \frac{x_a x_b}{C_f(m)}\right], \qquad C_f(m) = \frac{m(N-m)(N+1)}{N}. \tag{16}$$

Here $\omega_a = C_a/C_m$, $\sum \omega_a = 1$. The neutral Casimir identity in (16) was proved by trace covariance in the earlier force-norm note; it does not use a charged exchange. Its framework is the usual irreducible embedded-ensemble covariance [1, 2].

For all $a, b \le p$ on the central window one has

$$|f_{ab}(m)| \le K, \qquad |f_{ab}(m) - f_{pp}(m)| \le K(u+v)/p, \tag{17}$$

$$|f_{ab}(m-1) - f_{ab}(m)| \le Kp/N^2. \tag{18}$$

These bounds follow directly from the terminating neutral series. Its terms c_j satisfy $|c_j| \le K_0^j/j!$ and exactly

$$\frac{c_j(m-1)}{c_j(m)} = \frac{m(n-j)}{n(m-j)}, \quad \frac{c_j(m-1) - c_j(m)}{c_j(m)} = \frac{j(n-m)}{n(m-j)}. \tag{19}$$

Thus $|n-m| \le Kp$ proves the crossing part of (18). The C_f part uses $C_f(m) - C_f(m-1) = (n-m)(N+1)/N$. For (17), decreasing a by one multiplies a nonzero series term by $(a-j)(N+2-a)/[a(N+2-a-j)]$. For $a \ge p/2$ its difference from 1 is at most Kj/p; sum and telescope. If either rank is below $p/2$, boundedness already gives (17). The polynomial prefactors obey the same bound.

The exact normalized weight ratio is

$$\zeta_a = \frac{\omega_a^{(m-1)}}{\omega_a^{(m)}} = \frac{(m-p)n^2}{(n+p)(mn-x_a)}, \qquad \zeta_a - 1 = \frac{(n+p)x_a - pn(N+1)}{(n+p)(mn-x_a)}. \tag{20}$$

At $a = p$ the last numerator is $p^2(m-p)$; changing a to $p-u$ changes it by at most KuN^2. Hence $|\zeta_a - 1| \le K(1+u)/N$ and $\zeta_a \le K$. Subtract $f_{pp}(m)$ in the change of probability weights, using their equal total mass. Equations (17)–(18) and the uniform factorial input tail then give

$$|F_{m-1} - F_m| \le \frac{K}{Np}\mathbb{E}_{\omega\otimes\omega}[(1+u+v)(u+v)] + \frac{Kp}{N^2} \le \frac{K}{Np}. \tag{21}$$

This normalization cancellation is essential before multiplication by $h_0 = O(N)$.

4 The cross limit from the three polarization terms

In the first term of (11) separate $t_r = h_0 + \delta_r$. Since $\mathcal{I}_m$ is isometric and preserves W, its constant part is $h_0(F_{m-1} - F_m)/2 = O(p^{-1})$ by (21). The remaining part also vanishes. Indeed

$$\begin{aligned}\left|\sum W\delta_r(\|V\|^2 - \|U\|^2)\right| &\le \left(\sum (W\delta_r)^2\|V+U\|^2\right)^{1/2}\left(\sum \|E\|^2\right)^{1/2}\\ &\le K/N. \end{aligned} \tag{22}$$

The first factor is bounded by the fourth output moment and mixed input moments: $(W\delta_r)^2 \le KS^2(1+u+v+S)^2$. The sixth-moment bound provides more than is needed here.

For the last term, $W \le Kp$, $t_r \le KN$, and (13) imply

$$0 \le \frac{1}{2}\sum W t_r \|E\|^2 \le Kp/N = O(p^{-1}). \tag{23}$$

It remains to evaluate the middle term. Equations (14) and $W = S[1+O(p/N)]$, with the weighted moment bounds, give

$$\begin{aligned}\sum W(t_r - h_0)\|U\|^2 &= \sum S(u+v+S-2\lambda_N)\|U\|^2 + O(p^{-1})\\ &= \frac{\|\mathcal{S}\|^2 + \langle \mathcal{S}, L_{\mathrm{in}}\rangle + (2\bar{u} - 2\lambda_N)\langle \mathcal{S}, B\rangle}{C_m^2} + O(p^{-1}).\end{aligned} \tag{24}$$

The previously proved central product-transport theorem gives $\|\mathcal{S}\|^2/C_m^2 \to s$, $\langle \mathcal{S}, L_{\mathrm{in}}\rangle/C_m^2 \to 0$, $\langle \mathcal{S}, B\rangle/C_m^2 \to b$, and $\bar{u} - \lambda_N \to 0$, all uniformly in the window. Combining these three terms proves

$$\frac{\langle \mathcal{X}/N, O\rangle}{C_m^2} \longrightarrow s. \tag{25}$$

Finally $\|\mathcal{X}/N - \mathcal{S}\|/C_m = O(p^{-1})$ and $\|O\|/C_m \le K$ replace the proxy, proving (6).

The finite candidate $g[(x_a + x_b)\xi - m(\xi - \chi)]$ for the charged exchanged Casimir is not an input to this argument. Its exact general-charge operator derivation remains open in the earlier note. The central asymptotic cross is now established independently of it. In particular, we have not replaced its missing proof by a fit to finite-p data.

5 Projection and the quadratic memory core

Let

$$v = s - b^2/z = 2\lambda(1-q) + \frac{16\lambda^2 q}{1+q}, \qquad \Delta = a_* - s = 2\lambda(1+q). \tag{26}$$

Both are positive for $\lambda > 0$. Equation (6) has determinant $zlv\Delta > 0$. The residual after projecting $\mathcal{S}$ onto $\mathrm{span}(B, L_{\mathrm{in}}, O)$ is, to leading order,

$$\mathcal{T} = \mathcal{S} - \frac{v}{v+\Delta}O - \frac{b\Delta}{z(v+\Delta)}B, \qquad \frac{\|\mathcal{T}\|^2}{C_m^2} \to \frac{v\Delta}{v+\Delta} > 0. \tag{27}$$

Using the actual finite-N orthogonal projector has the same limit: its three-field Gram has a positive limiting inverse, uniformly on the stated compact parameter sets.

To connect with the actual memory coefficient, recall the raw core

$$\mathcal{E} = \sum_i [H,[H,c_i]]^\dagger [H,[H,c_i]] + [h,[h,A]]/4, \quad A = \mathrm{Rem}(H_{m-1}^2) + (2p-m)h^2. \tag{28}$$

Full-Fock operations precede the charge restriction. The earlier strong central expansion, with bounded physical C_m, is

$$\begin{aligned} \mathsf{W}_2\mathcal{E} &= C_m\left[2p(1-q_N)^2 + \overline{K}_N + 4\lambda_N x(1-q_N)^2\right]B \\ &\quad + C_m\gamma_N O + 8\lambda_N C_m(q_N^2 - q_N)L_{\mathrm{in}} + 8\lambda_N C_m q_N^2\mathcal{S} + o_{L^2}(1), \\ \gamma_N &= (3 - 4q_N + q_N^2)/2, \qquad q_N = e^{-4\lambda_N}, \\ \overline{K}_N &= \lambda_N(1 - 10q_N + 9q_N^2) - 16\lambda_N^2 q_N^2. \end{aligned} \tag{29}$$

Keep the actual finite parameters in the growing conserved term. Projecting onto the complement of (B, L_{in}, O) now yields

$$\frac{\|(1-\Pi_N)\mathsf{W}_2\mathcal{E}\|^2}{C_m^4} \longrightarrow 64\lambda^2 q^4 \frac{v\Delta}{v+\Delta}. \tag{30}$$

The charge-dependent correction multiplies B and is removed by this specified projection; it is not absent from the raw core. For $C_m \to J_s^2/\lambda$, first take the double-scaled limit, then $\lambda \downarrow 0$. The variance in (30) becomes

$$V_2(\lambda) = \frac{64 J_s^8 q^4}{\lambda^2}\frac{v\Delta}{v+\Delta} \longrightarrow 1024 J_s^8. \tag{31}$$

This is an equal-time second-chaos result. The complete raw memory spectrum also requires the other chaos and the energy descendants.

6 Checks, provenance and the next calculation

The proof uses the established neutral covariance, adjacent removal isometry, central coherent-force norm, and the preceding product-transport Gram. The new ingredient is the neutral drift estimate inside an exact polarization identity. It is an application of these tools, with no claim of priority over the embedded-ensemble literature.

The three new scientific records are:

- `neutral-drift.json`: 758 exact normalized weight ratios; four independent finite rational neutral moments; 36 complete-rank adjacent-sector differences at $\lambda = 0.1, 0.25, 0.5$, $p = 16, 32, 64, 128$, $x = -2, 0, 2$. All terminating Racah terms are included. The 80/120-digit refinement changes the reported quantities by at most $3.56\,10^{-74}$; comparisons use the numerical values before decimal serialization. The symbolic score is $N(\zeta_{p-u} - 1) \to 4(\lambda - u)$.

- `polarization-car.json`: complete Gaussian-coordinate and input/output-rank decompositions on $(N, p, m) = (5, 1, 3)$, $(6, 1, 3)$ and $(6, 1, 4)$. There are 3,217 ordered coordinate pairs across the three systems and six rank blocks per system. The maximal block polarization error is $2.78\,10^{-17}$. The source is the actual CAR force, not the candidate charged formula. Square-root and projected-coefficient checks use floating arithmetic with tolerance 10^{-12}.

- `gram-assembly.json`: exact symbolic projection, determinant, residual variance, core normalization, and ordered small-λ limit.

For example, at $\lambda = 0.5, x = 2$, the amplified neutral drift $h_0(F_{m-1} - F_m)/2$ is about -0.184286, -0.0987564, -0.0512180, -0.0260982 for $p = 16, 32, 64, 128$. The corresponding $Np(F_{m-1} - F_m)$ remains of order one; the finite correction is retained rather than replaced by zero. Uniformity is proved by the inequalities on page 3, not by the scan.

The next task is central-window energy insertions and the raw U_2 spectrum, followed by control outside fixed charge windows and two-insertion finite-memory remainders. The bulk meson space/mass dictionary is still to be constructed. The present result does not infer a new bulk particle from a positive norm.

References

[1] J. Tall and S. Tomsovic, *Double-scaled bosonic and fermionic embedded ensembles, complex SYK, and the dual Hilbert space*, arXiv:2604.14522v2, Sec. 2.1. Fixed-charge covariance is a prior input.

[2] A. Day and Chengheng, local notes *Central charge windows*, *Coherent memory norms on central charge windows*, and *Transporting quadratic products across central charge windows* (6 November 2026).